

OPTIMAL DESIGN FOR THE ELISA TYPE OF IMMUNOASSAYS: CLASSICAL CALIBRATION VERSUS PARALLEL FITTING

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Summary: Immunoassays such as the radio immunoassays (RIA) and the enzyme-linked immunosorbent assays (ELISA) are techniques for quantifying biochemical substances in specific media. The commonly used designs for these assays are those for classical calibration although this method has not been proven to be the best for the assays. One competing alternative calibration method that is rarely used in immunoassays is the parallel fitting method. In this paper we use the ELISA example to show that the classical calibration method is inaccurate and imprecise for predicting extreme unknown concentrations of biochemical substances whereas the parallel fitting method is reasonably accurate and precise for predicting both moderate and extreme unknown concentrations of the substances. Furthermore, we show how to extend optimal designs for classical calibration to optimal designs for parallel fitting. The conclusion of this paper is that the parallel fitting method should be preferred over the classical calibration method if in the area of application of the immunoassay both moderate and extreme unknown concentrations of the biochemical substance in the medium are of interest.

1. Introduction

The enzyme-linked immunosorbent assay (ELISA) is described in Roche and Jones (1997) and in Bunch, Roche and Harrison (1990) as a technique for detecting and/or quantifying biochemical substances in specific media, e.g. antibodies, hormones and drugs in body fluids; toxins in foods; and agricultural chemicals and hazardous wastes in environmental samples. Basically, the ELISA technique works as follows.

The biochemical analyses of the samples with known (z) and unknown (z^*) $\log$ concentrations of the substance yield coloured products with respective optical densities y and y^* which are related to the corresponding $\log$ concentrations of the substance by the parallel four-parameter logistic regression models:

$$y = \phi_4 + \frac{\phi_1}{1 + \exp\{-\phi_2(z - \phi_3)\}} + \sigma_z \epsilon, \quad E[\epsilon] = 0, \quad Var[\epsilon] = \sigma^2, \quad (1)$$

$$y^* = \phi_4 + \frac{\phi_1}{1 + \exp\{-\phi_2(z^* - \phi_3)\}} + \sigma_{z^*} \epsilon^*, \quad E[\epsilon^*] = 0, \quad Var[\epsilon^*] = \sigma^2, \quad (2)$$

where the ϵ and the ϵ^* are independent and identically distributed random errors, σ_z^2 and $\sigma_{z^*}^2$ are known variance functions, and $\phi_1 > 0$, $|\phi_2| \neq 0$, $|\phi_3| < \infty$ and $\phi_4 \geq 0$ are the regression parameters. Then the regression methods such as the classical calibration method (Osborne, 1991) and the parallel fitting method (Finney, 1978, pp. 40-41, Ndlovu and Preater, 2001) are used to predict the unknown concentrations of the substance (z^*) in the samples.

U	S	S	S	S	S	S	U	S	S	S	S
S	S	U	S	S	U	S	S	S	S	S	S
U	S	S	S	S	U	S	S	S	S	S	U
U	S	S	S	S	U	S	S	S	U	U	S
S	S	S	U	S	S	U	S	S	S	U	S
S	S	S	S	U	S	U	S	S	S	U	S
U	S	S	S	S	S	S	U	S	S	S	S
S	S	S	U	S	U	U	S	U	S	S	S

Figure 1 A 96-well microplate for the ELISA: U are wells for the unknowns, and S are wells for the standards

The design of the ELISA is as follows. Samples with known and unknown concentrations of the substance are analyzed in wells of a fixed K -well microplate (Figure 1) that is divided into wells for the *standards* and wells for the *unknowns*. The standards are k samples ($k \geq 4$) with different but known *log* concentrations (z) of the substance. Each standard is analyzed in n_i ($i = 1, 2, \dots, k$) of the K wells to obtain the standard sample $S = \{y_{ij}, z_i\}_{j=1, i=1}^{n_i, k}$ for fitting the regression model. The unknowns are k^* individual samples with unknown *log* concentrations (z^*) of the substance. Each of the k^* unknowns is analyzed as is in n^* of the remaining $k^*n^* = K - N$ ($N = \sum_{i=1}^k n_i$) wells if the unknowns are to be predicted using the classical method, or, in series of d dilutions/concentrations of each unknown if the unknowns are to be predicted using the parallel fitting method. When the latter method is adopted, d dilutions/concentrations of each unknown are prepared using known *log* dilution/concentration (LDC) factors α_i , $i = 1, 2, \dots, d$. The LDC factors are chosen to maximize the chance that at least one $z_i^* = z^* + \alpha_i$ is in the region of the unknowns (region L in Figure 2) where they can be predicted with maximum possible accuracy (minimum absolute or squared bias) and precision (minimum variance). This region is the

neighbourhood of the parameter ϕ_3 . Each of the d dilutions/concentrations is analyzed in m of the $n^* = dm$ wells. This analysis yields the unknown samples $U = \{y_{ij}^*, z_i^*\}_{j=1, i=1}^{m, d}$. Thus, the exact design for predicting the unknowns using the parallel fitting method is given by the array:

$$\xi_K^p = \left\{ \begin{array}{cccc|cc} z_1 & z_2 & \dots & z_k & S & U \\ n_1 & n_2 & \dots & n_k & N & K-N \end{array} \middle| \begin{array}{cccc} \alpha_1 & \alpha_2 & \dots & \alpha_d \\ m & m & \dots & m \end{array} \right\} := \{\xi_K^z | \xi_K^c | \alpha_K\} \quad (3)$$

If for K large enough,

$$N = (1 - k^*q)K, \quad q \in (1/K, 1/k^*), \quad n_i = p_i N, \quad p_i \in [0, 1), \quad \sum_{i=1}^k p_i = 1,$$

then the continuous counterpart (Silvey, 1980, p. 13) of design ξ_K^p is given by:

$$\xi^p = \left\{ \begin{array}{cccc|cc} z_1 & z_2 & \dots & z_k & S & U \\ p_1 & p_2 & \dots & p_k & 1 - k^*q & k^*q \end{array} \middle| \begin{array}{cccc} \alpha_1 & \alpha_2 & \dots & \alpha_d \\ \frac{1}{d} & \frac{1}{d} & \dots & \frac{1}{d} \end{array} \right\} := \{\xi^z | \xi^c | \alpha\} \quad (4)$$

In design ξ^p , ξ^c is a design measure which allocates a proportion $(1 - k^*q)$ of the K wells to the standards (S) and equally divides the remainder among the k^* unknowns (U), ξ^z is a design measure which allocates a proportion p_i of the wells assigned to a particular standard z_i , and α is a design measure which equally divides the wells of each unknown among its d dilutions/concentrations.

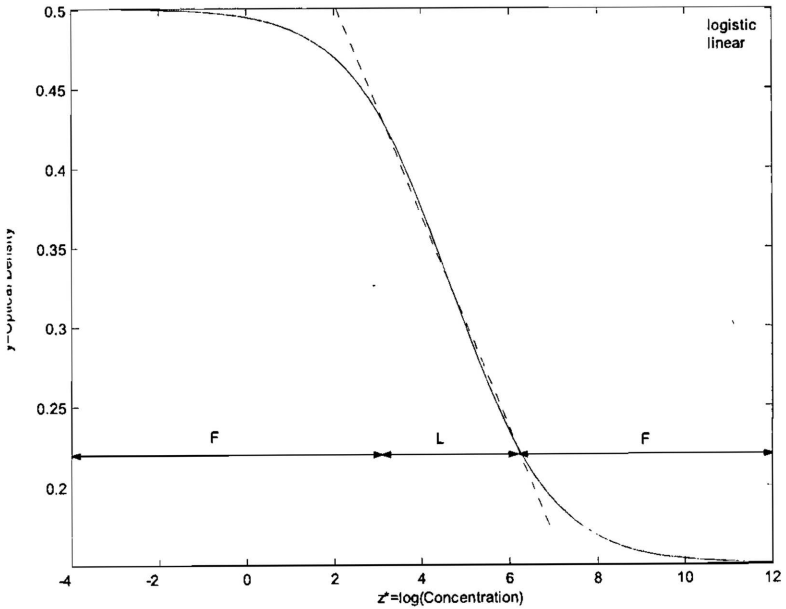


Figure 2 The mean response function of the four-parameter logistic model (2) for $(\phi_1, \phi_2, \phi_3, \phi_4) = (.350, -.872, 4.662, .151)$

The classical calibration method is the usual method for predicting the unknowns although the method is only efficient for predicting moderate unknowns (in region L of Figure 2) as shall be demonstrated in section 2 of this paper. The method is inaccurate and imprecise for predicting extreme unknowns because the mean response function $E[y|z]$ is nearly flat at the tails (regions F in Figure 2). On the other hand Ndlovu and Preater (2001) report that the parallel fitting method can predict a wider range of the unknowns, with reasonable accuracy and precision, than the classical calibration method if the design for parallel fitting is optimally chosen. This report suggests that the parallel fitting method should be preferred over the classical method if in

the areas of application of the ELISA (such as mentioned at the beginning of this section) it is desirable to predict both moderate and extreme unknown concentrations of the substances in the media.

The objectives of this paper are: (1) to compare the accuracy and the precision of the classical calibration and the parallel fitting methods in predicting both moderate and extreme unknowns, and (2) to show how to construct design ξ^P (parallel fitting) for predicting the widest possible range of the unknowns with maximum possible accuracy (minimum absolute or squared bias) and precision (minimum variance). The latest literature on the designs for the ELISA appears to be papers by Ndlovu and Rosenberger (2006), Rocke and Jones (1997) and Bunch, Rocke and Harrison (1990). These authors discuss the designs for predicting the unknowns using the classical calibration method. As for the designs for parallel fitting, the latest literature appears to be a paper by Ndlovu and Preater (2001) which discusses the construction of the designs for calibration using a piecewise simple linear regression model. For this model, the authors report that the parallel fitting method is more robust and reasonably efficient for predicting both moderate and extreme unknowns than the classical calibration method if the design for parallel fitting is optimally chosen.

This paper does not propose new optimality criteria for the design for parallel fitting but generalises the schemes I and II of Ndlovu and Preater (2001) to choosing the optimal *log* dilution/concentration (LDC) factors in the design for the ELISA (α in ξ^P). That is, schemes I and II of Ndlovu and Preater (2001) are found to be special cases of the design approach in this paper. The paper is organized as follows: The parallel fitting and the classical calibration methods are described and compared in section 2. This section demonstrates that the classical calibration method is only more accurate and

more precise than the parallel method for predicting moderate unknowns but is less accurate and less precise for predicting extreme unknowns. Section 3 discusses the construction of the designs for parallel fitting in general and the construction of the designs for the ELISA in particular. This section also evaluates the performance of the designs for the ELISA, and reevaluates the performance of the parallel fitting method relative to that of the classical method. Section 4 is the conclusion.

2. Parallel fitting and classical calibration methods

The parallel fitting method simultaneously estimates the unknown *log* concentration z^* along with the model (1) parameters from both S and U using estimation methods such as the maximum likelihood method (ML) and the iteratively reweighted least squares (IRLS) method. The IRLS estimate of $(\phi_1, \phi_2, \phi_3, \phi_4, z^*)$ is $(\hat{\phi}_1, \hat{\phi}_2, \hat{\phi}_3, \hat{\phi}_4, \hat{z}^*)$ which minimizes:

$$Q = \sum_{i=1}^k \sum_{j=1}^{n_i} \left(\frac{y_{ij} - \phi_4 - \phi_1 \eta_i}{\sigma_i} \right)^2 + \sum_{i=1}^d \sum_{j=1}^{m_i} \left(\frac{y_{ij}^* - \phi_4 - \phi_1 \eta_i^*}{\sigma_i^*} \right)^2 \quad (5)$$

where

$$\eta_i = \frac{1}{1 + \exp\{-\phi_2(z_i - \phi_3)\}}, \quad \eta_i^* = \frac{1}{1 + \exp\{-\phi_2(z_i^* - \phi_3)\}}, \quad z_i^* = z^* + \alpha_i,$$

and for the ELISA, Rocke and Jones (1997) report that $\sigma_i^2 = (E[y_{ij}|z_i])^2$ and $\sigma_i^{*2} = (E[y_{ij}^*|z_i^*])^2$. An initial estimate of z^* for starting iterative algorithms for minimizing (5) can be $\tilde{z}^*$ which minimizes the second term of (5) evaluated at $(\phi_1, \phi_2, \phi_3, \phi_4) = (\tilde{\phi}_1, \tilde{\phi}_2, \tilde{\phi}_3, \tilde{\phi}_4)$, the IRLS estimate of $(\phi_1, \phi_2, \phi_3, \phi_4)$ from S only. It is noted that if U does not contain any information about $(\phi_1, \phi_2, \phi_3, \phi_4)$, then $\tilde{z}^*$ is an estimate of z^* which may be as good as $\hat{z}^*$. The asymptotic moment matrix of $(\hat{\phi}_1, \hat{\phi}_2, \hat{\phi}_3, \hat{\phi}_4, \hat{z}^*)$ is

given by:

$$\Omega(\phi, z^*, \xi^p) := \frac{K}{\sigma^2} \begin{bmatrix} (1 - k^*q)I + qI^* & q\gamma \\ q\gamma & q\tau \end{bmatrix}_{5 \times 5} \quad (6)$$

where

$$I := \left[\sum_{i=1}^k \frac{p_i}{\sigma_i^2} \frac{\partial E[y_{ij}]}{\partial \phi} \frac{\partial E[y_{ij}]}{\partial \phi} \right]_{4 \times 4}, \quad I^* := \left[\sum_{j=1}^d \frac{1}{d\sigma_i^{*2}} \frac{\partial E[y_{ij}^*]}{\partial \phi} \frac{\partial E[y_{ij}^*]}{\partial \phi} \right]_{4 \times 4},$$

$$\gamma := \left(\sum_{i=1}^d \frac{1}{d\sigma_i^{*2}} \frac{\partial E[y_{ij}^*]}{\partial z^*} \frac{\partial E[y_{ij}^*]}{\partial \phi} \right)_{4 \times 1} \quad \text{and} \quad \tau = \sum_{i=1}^d \frac{1}{d\sigma_i^{*2}} \left(\frac{\partial E[y_{ij}^*]}{\partial z^*} \right)^2.$$

When all the inverses exist, the asymptotic variance of $\hat{z}^*$ is given by:

$$\begin{aligned} V(\phi, z^*, \xi^p) &:= \text{the } (5, 5)^{th} \text{ entry of } \Omega(\phi, z^*, \xi^p) \\ &= \frac{\sigma^2}{K} \left\{ \frac{1}{\tau q} + \frac{\gamma^t}{\tau} [(1 - k^*q)I + qI^* - q\gamma\tau^{-1}\gamma^t]^{-1} \frac{\gamma}{\tau} \right\} \quad (7) \\ &= v_1 + v_2 \end{aligned}$$

In (7) the first (v_1) and the second (v_2) terms in the sum are the respective components of variance due to the errors in the standard and unknown samples.

Figure 3 displays the graph of the asymptotic variance (7), for the ELISA, evaluated at design

$$\xi^p = \left\{ \begin{array}{cccc|cc|ccc} -\infty & 3.974 & 6.494 & \infty & S & U & -3.021 & 0 & 3.021 \\ .25 & .25 & .25 & .25 & .25 & .75 & 1/3 & 1/3 & 1/3 \end{array} \right\} \quad (8)$$

and $(\phi_1, \phi_2, \phi_3, \phi_4, \sigma, k^*, K) = (.35, -.872, 4.662, .151, .045, 12, 96)$, versus the unknown log concentration z^* . (With design (8), each of the 4 standards and each of the 12 unknowns is allocated 6 wells of the 96-well

microplate. Furthermore, each of the 3 dilutions and concentrations of each unknown is allocated 2 of the 6 wells allocated the unknown.)

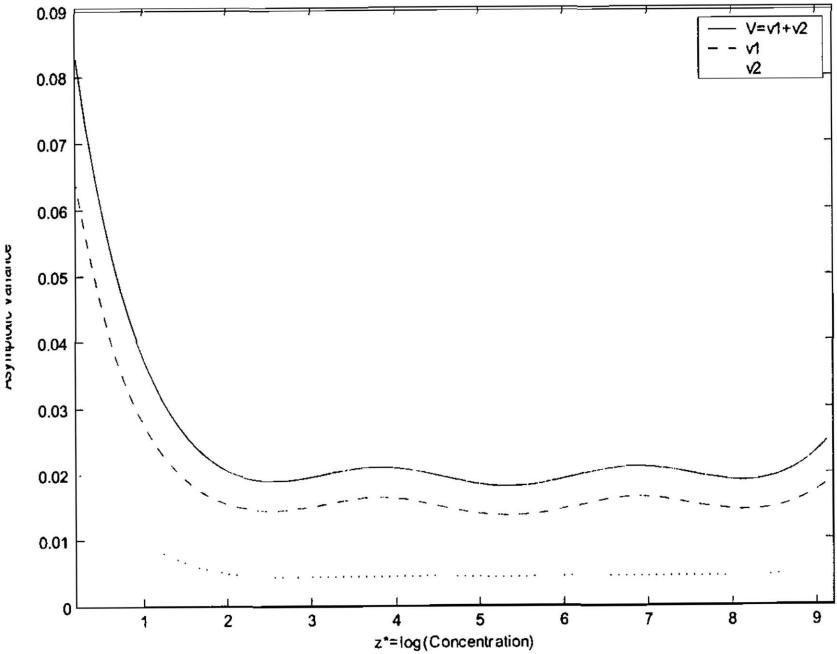


Figure 3 The asymptotic variance of $\hat{z}^*$ (ELISA and parallel fitting) versus z^* : $V = V(\phi, z^*, \xi^p)$ given by (7); $v1 = v_1$ and $v2 = v_2$ are the first and second terms (components of variance) in (7);
 $(\phi_1, \phi_2, \phi_3, \phi_4, \sigma, k^*, K) = (.35, -.872, 4.662, .151, .045, 12, 96)$

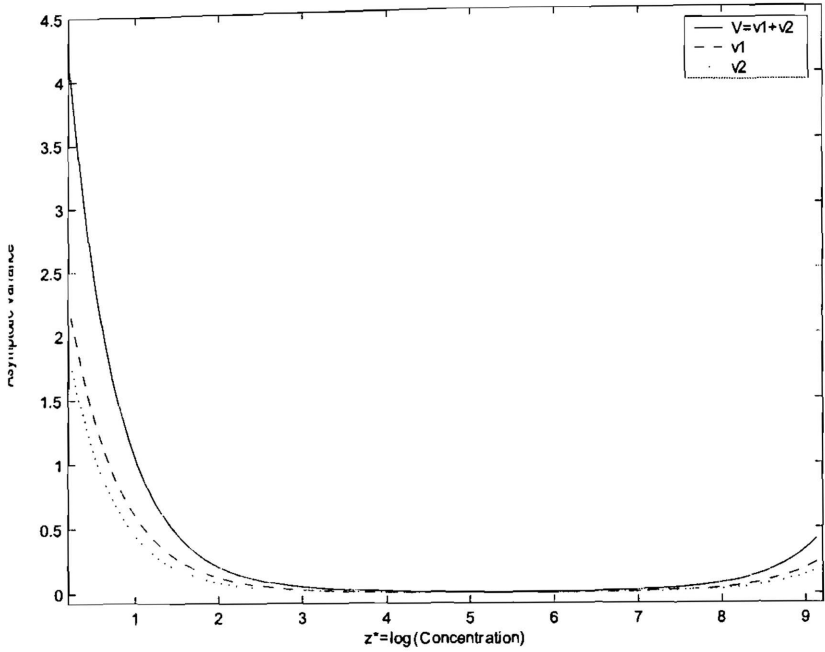


Figure 4 The asymptotic variance of $\hat{z}^*$ (ELISA and classical calibration) versus z^* : $V = V(\phi, z^*, \xi)$ given by (13); $v_1 = v_1$ and $v_2 = v_2$ are the first and second terms (components of variance) in (13);
 $(\phi_1, \phi_2, \phi_3, \phi_4, \sigma, k^*, K) = (.35, -.872, 4.662, .151, .045, 12, 96)$

The graph shows that neither component of the design (ξ^z and ξ^c) and the variance (7) can be ignored when designing the ELISA. Furthermore, both components of variance $\rightarrow \infty$ as $z^* \rightarrow \pm\infty$ due to the fact that for the ELISA,

$$\sigma_i^{*2} = (\phi_4 + \phi_1 \eta_i^*)^2, \quad \eta_i^* = \frac{1}{1 + \exp\{-\phi_2(z_i^* - \phi_3)\}}, \quad z_i^* = z^* + a_i, \quad (9)$$

and hence both components of variance are dominated by the inverse of

$$\tau = \sum_{i=1}^d \frac{\phi_1^2 \phi_2^2 \eta_i^{*2} (1 - \eta_i^*)^2}{d(\phi_4 + \phi_1 \eta_i^*)^2} \rightarrow 0 \text{ as } \eta_i^* \rightarrow 0 \text{ or } 1 \text{ as } z^* \rightarrow \pm\infty \quad (10)$$

for all finite α_i . It is clear from the expression for τ that if for some z^* at least one z_i^* is in the neighbourhood of ϕ_3 then

$$0 \ll \tau \ll \frac{\phi_1^2 \phi_2^2}{(4\phi_4 + 2\phi_1)^2} \quad (11)$$

Hence, in Figure 3, the variance (7) is smallest for those z^* for which τ is closest to its upper limit, i.e. for those z^* for which at least one z_i^* is in the neighbourhood of ϕ_3 . The upper limit of τ is not achievable if at least two α'_i 's are different as is the case in the design for parallel fitting.

The parallel fitting method is the classical calibration method if the component α of ξ^p (4) is degenerate at $\mathbf{0}$. In this case (4) becomes $\xi^p = \{\xi^z | \xi^c\} := \xi$, the design for classical calibration. Furthermore, the classical calibration estimate of z^* is

$$\hat{z}^* = \hat{\phi}_3 + \frac{1}{\hat{\phi}_2} \ln \left\{ \frac{\hat{\eta}^*}{1 - \hat{\eta}^*} \right\}, \quad \hat{\eta}^* = \frac{\bar{y}^* - \hat{\phi}_4}{\hat{\phi}_1}, \quad \bar{y}^* = \frac{1}{dm} \sum_{i=1}^d \sum_{j=1}^m y_{ij}^* > \hat{\phi}_4 \quad (12)$$

where $(\hat{\phi}_1, \hat{\phi}_2, \hat{\phi}_3, \hat{\phi}_4)$ is $(\phi_1, \phi_2, \phi_3, \phi_4)$ which minimize the first term of (5). The asymptotic variance of $\hat{z}^*$ is given by:

$$\begin{aligned} V(\phi, z^*, \xi^p) &= \frac{\sigma^2}{K} \left\{ \frac{1}{\tau q} + \frac{\gamma^t}{\tau} [(1 - k^* q)I]^{-1} \frac{\gamma}{\tau} \right\} \\ &= v_1 + v_2 := V(\phi, z^*, \xi) \end{aligned} \quad (13)$$

upon evaluating (7) at ξ^p with its α component degenerate at $\mathbf{0}$ (in which case $I^* = \gamma \tau^{-1} \gamma$).

Figure 4 displays the graph of the variance (13) (and its components v_1 and v_2) versus z^* , where ξ is design (8) with the component α degenerate at 0. Compared with Figure 3, Figure 4 shows that the variance (7) $\rightarrow \infty$ faster than the variance (7) as $z^* \rightarrow \pm\infty$. This is because in this case

$$\sigma^{*2} = (\phi_4 + \phi_1\eta^*)^2, \quad \eta^* = \frac{1}{1 + \exp\{-\phi_2(z^* - \phi_3)\}},$$

and hence

$$\tau = \frac{\phi_1^2 \phi_2^2 \eta^{*2} (1 - \eta^*)^2}{(\phi_4 + \phi_1\eta^*)^2} \rightarrow 0$$

faster than the τ given by (10) as $\eta^* \rightarrow 0$ or 1 as $z^* \rightarrow \pm\infty$. In fact for classical calibration (compare with (11))

$$0 \leq \tau \leq \frac{\phi_1^2 \phi_2^2}{(4\phi_4 + 2\phi_1)^2} \quad (14)$$

which suggests that the parallel fitting method is more robust for predicting extreme z^* than the classical calibration method. What is not clear from Figures 3 and 4 is that variance (13) is smaller than the variance (7) for $z^* \in (4, 7)$. The reason for this is that for $z^* = \phi_3$, τ (classical calibration) is equal to $\phi_1^2 \phi_2^2 / (4\phi_4 + 2\phi_1)^2$ which is the upper limit of τ (parallel fitting) given by (11). This fact suggests that the classical calibration method is more efficient for predicting z^* in the neighbourhood of ϕ_3 than the parallel fitting method.

Consider the bias of the IRLS estimate of $(\phi_1, \phi_2, \phi_3, \phi_4, z^*)' := \phi^*$, $(\hat{\phi}_1, \hat{\phi}_2, \hat{\phi}_3, \hat{\phi}_4, \hat{z}^*)$, by the parallel fitting method, that is induced by the nonlinearity of the models for the standards and the unknowns. Following

Box (1971) it can be deduced that the bias of the IRLS estimate is given by:

$$\begin{aligned}
 \mathbf{B}(\phi^*, \xi^p) := & -\frac{K}{2} \Omega^{-1}(\phi, z^*, \xi^p) \times \\
 & \left\{ (1 - k^* q) \sum_{i=1}^k \frac{p_i}{\sigma_i^2} \frac{\partial E[y_{ij}]}{\partial \phi^*} \times \right. \\
 & \quad \text{Trace} \left(\Omega^{-1}(\phi, z^*, \xi^p) \frac{\partial^2 E[y_{ij}^*]}{\partial \phi^{*t} \partial \phi} \right) \\
 & + q \sum_{i=1}^d \frac{1}{d\sigma_i^{*2}} \frac{\partial E[y_{ij}]}{\partial \phi^*} \times \\
 & \quad \left. \text{Trace} \left(\Omega^{-1}(\phi, z^*, \xi^p) \frac{\partial^2 E[y_{ij}^*]}{\partial \phi^{*t} \partial \phi} \right) \right\} \\
 = & \mathbf{b}_2 + \mathbf{b}_1
 \end{aligned} \tag{15}$$

where $\mathbf{b}_1$ and $\mathbf{b}_2$ are the respective components of bias due to the nonlinearity of the models for the standards and the unknowns.

The bias of $\hat{z}^*$ is given by:

$$B(\phi, \xi^p) := \text{the } 5^{th} \text{ component of } \mathbf{B}(\phi^*, \xi^p) = b_1 + b_2 \tag{16}$$

where b_1 and b_2 are the 5th components of $\mathbf{b}_1$ and $\mathbf{b}_2$ (15), respectively.

Figure 5 displays the graph of the bias (16) and its components b_1 and b_2 (evaluated at design (8)) versus z^* , and Figure 6 displays the corresponding graph of the bias of the classical calibration estimator of z^* . As with the variance (Figures 3 and 4), Figures 5 and 6 show that the bias of the classical calibration estimator $\rightarrow \pm\infty$ faster than the bias of the parallel fitting method estimator of z^* as $z^* \rightarrow \pm\infty$. Furthermore, the absolute bias of the parallel fitting estimator is smaller than the absolute bias of the classical calibration estimator except when $z^* \in (4.83, 5.83)$.

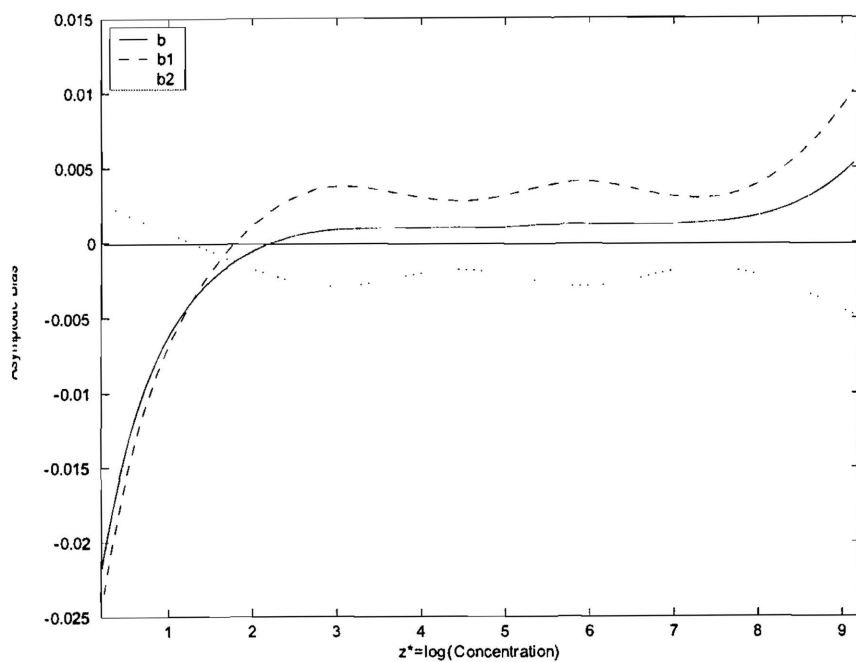


Figure 5 The asymptotic bias of $\hat{z}^*$ (ELISA and parallel fitting) versus z^* : $B = B(\phi, \xi^p)$ given by (16); $b1 = b_1$ and $b2 = b_2$ are the first and second

terms (components of variance) in (16);
 $(\phi_1, \phi_2, \phi_3, \phi_4, \sigma, k^*, K) = (.35, -.872, 4.662, .151, .045, 12, 96)$

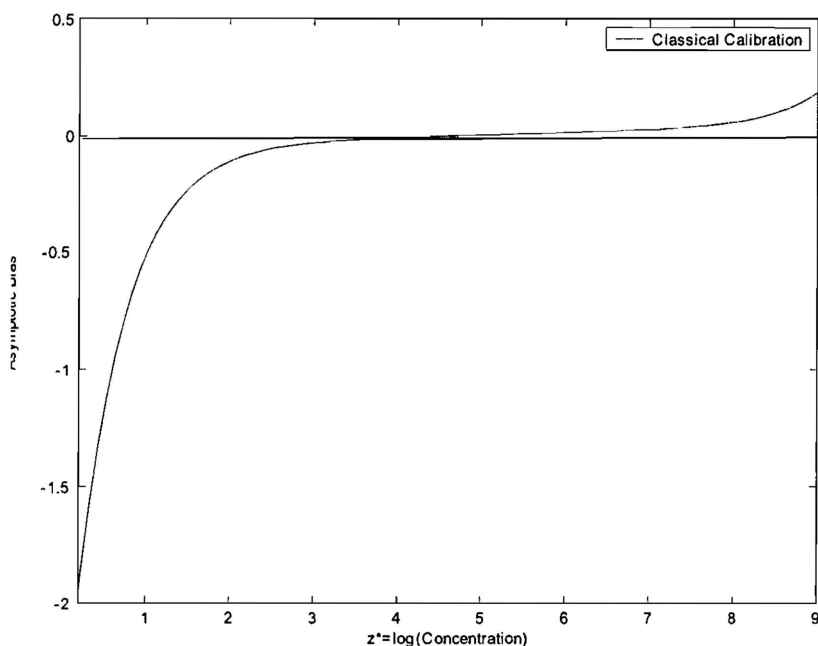


Figure 6 The asymptotic bias of $\hat{z}^*$ (ELISA and classical calibration) evaluated at design (8) with the α component degenerate at 0 and $(\phi_1, \phi_2, \phi_3, \phi_4, \sigma, k^*, K) = (.35, -.872, 4.662, .151, .045, 12, 96)$

3. Optimal design for parallel fitting

The numerical results of section 2 suggest that the parallel fitting method should be preferred over the classical method if in the area of application of the ELISAs it is desired to predict both moderate and extreme unknowns. This section discusses the construction of the optimal design for parallel fitting.

An ideal design for parallel fitting (ξ^p given by (4)) is one that is robust to misspecification of the values of the model parameters and enables the prediction of moderate and extreme unknowns with the maximum possible accuracy and precision. Here, the measure of accuracy is the bias or the squared bias of $\hat{z}^*$, and the measure of precision is the asymptotic variance of $\hat{z}^*$ given by (7). As in Ford, Titterton and Kitsos (1989), and Chaloner and Larntz (1989) criteria such as the minimax and the expectation/Bayesian criteria may be used to define “near” ideal designs for parallel fitting. However, without any restrictions on the ξ^z and the α components of design ξ^p , designs ξ^p which minimize convex functions of (7) may be singular due to the fact that ξ^z and α have the same support. Ndlovu and Preater (2001) demonstrate this singularity using the simple linear regression model. The authors propose circumventing this problem by restricting ξ^z to the optimal design for classical calibration, and then optimizing α . This design approach is justified by the equivalence of the parallel fitting and the classical calibration methods when α is degenerate at 0 (see section 2).

Following Ndlovu and Preater (2001), the design for parallel fitting can be constructed in two stages as follows:

1. Construct an optimal design $\xi = \{\xi^z | \xi^c\}$ for classical calibration.
2. Restrict the ξ^z component of design ξ^p to the optimal design for classical calibration and then optimize both ξ^c and α .

This next section is focused on optimizing α given ξ^z and ξ^c .

3.1 Optimal log dilution/concentration (LDC) factors α

At stage 2 of the construction of the design for parallel fitting, the optimality criterion for choosing α is to choose one that *maximizes the chance that at least one $z_i^* = z^* + \alpha_i$ ($i = 1, 2, \dots, d$) is in the region of the unknowns* ($L := [L_-, L_+]$) *where they can be predicted with maximum possible accuracy*

(minimum absolute or squared bias) and precision (minimum variance). Figures 3 and 5 in section 2 show that it is practically impossible to accurately and precisely predict all the unknowns in $(-\infty, \infty)$ with any design ξ^p because the squared bias and the variance of $\hat{z}^* \rightarrow \infty$ as $z^* \rightarrow \pm\infty$ for any design ξ^p . Hence there is need to define a finite region, $F := [F_-, F_+] \supseteq L$, of the unknowns to be predicted.

Given L, F and a $\lambda \in [0, 1]$, then an optimal α satisfies:

$$\begin{aligned} F_- &= L_- - \alpha_1 \implies \alpha_1 = L_- - F_- \\ \alpha_{i+1} &= (1 - \lambda)(L_- - L_+) + \alpha_i, \quad i = 1, 2, \dots, d-1 \\ F_+ &= L_+ - \alpha_d \implies \alpha_d = L_+ - F_+ \end{aligned} \quad (17)$$

The equations are obtained by noting that $z_i^* \in L \iff z^* \in [L_- - \alpha_i, L_+ - \alpha_i]$. Then F is partitioned into overlapping intervals of the form $[L_- - \alpha_i, L_+ - \alpha_i]$, and in such a way that the lower end point of the $(i+1)^{st}$ interval is a convex combination (with mixing constant λ) of the end points of the i^{th} interval. λ indicates the degree of the overlap of the intervals, and hence implicitly determines the maximum number of the $z_i^* \in L$ for each $z^* \in F$. Solving the equations (17) for the α_i obtains:

$$\begin{aligned} \alpha_1 &= L_- - F_- \\ \alpha_i &= (i-1)(1-\lambda)(L_- - L_+) + (L_- - F_-), \quad i = 2, \dots, d \\ (F_+ - F_-) &= (d - \lambda d + \lambda)(L_+ - L_-) \end{aligned} \quad (18)$$

The second equation of (18) shows that F, L, d and λ cannot be chosen arbitrarily. For example, if $\lambda = 1$ then $F = L$ and the $\alpha_i = 0$. On the other hand, if $\lambda = 0$, then $(F_+ - F_-) = d(L_+ - L_-)$ and the d intervals $[L_- - \alpha_i, L_+ - \alpha_i]$ partition F . In practice L will be fixed by the nature of the mean response function, hence, subject to $d \leq qK$ (see design (3)), the possible values of (d, λ) will be specified by choice of F , or, the size of F will be specified by choice of (d, λ) . In the absence of any prior knowledge about the location of the unknowns (as will usually be the case) F can be

chosen symmetric about L . In this case F is given by:

$$\begin{aligned} F_- &= \{(d - \lambda d + \lambda + 1)L_- - (d - \lambda d + \lambda - 1)L_+\} / 2 \\ F_+ &= \{(d - \lambda d + \lambda + 1)L_+ - (d - \lambda d + \lambda - 1)L_-\} / 2 \end{aligned} \quad (19)$$

Equations (18) and (19) are generalizations of schemes I ($\lambda = 0$) and II ($\lambda = 1/2$) of Ndlovu and Preater (2001).

In order to avoid explicitly specifying λ in (d, λ) , this paper considers the case when $L, d - \lambda d + \lambda = c > 1$ and $d \leq qK$ are user specified. In this case c specifies the size of F relative to L , and (c, d) implicitly specifies that $(1 - \lambda) = (c - 1)/(d - 1)$, and hence subject to $1 < c \leq d \leq qK$,

$$\begin{aligned} F_- &= \{(c + 1)L_- - (c - 1)L_+\} / 2 \\ F_+ &= \{(c + 1)L_+ - (c - 1)L_-\} / 2 \\ \alpha_i &= \left(\frac{1}{2} - \frac{i-1}{d-1}\right) (c - 1)(L_+ - L_-), \quad i = 1, 2, \dots, d \end{aligned} \quad (20)$$

We evaluate the performance of (20) in section 3.2 using an ELISA example.

3.2 ELISA example

A natural choice of the region L for the ELISAs is given by:

$$L = \phi_3 \pm \frac{1}{|\phi_2|} \ln \left\{ \frac{3 + \sqrt{3}}{3 - \sqrt{3}} \right\} \quad (21)$$

e.g. L in Figure 2. The limits of L are solutions of the equation $\frac{\partial^3 E[y]}{\partial z^3} = 0$.

With this choice of L the equations in (20) obtain:

$$\begin{aligned} F &= \phi_3 \pm \frac{c}{|\phi_2|} \ln \left\{ \frac{3 + \sqrt{3}}{3 - \sqrt{3}} \right\} \\ \alpha_i &= \left(\frac{1}{2} - \frac{i-1}{d-1}\right) (c - 1) \frac{2}{|\phi_2|} \ln \left\{ \frac{3 + \sqrt{3}}{3 - \sqrt{3}} \right\}, \quad i = 1, 2, \dots, d \end{aligned} \quad (22)$$

F , L and α can be estimated using a prior values of (ϕ_2, ϕ_3) and (c, d) . Table 1 displays the estimates of these quantities for $(\phi_2, \phi_3) = (-.872, 4.662)$ and various values of (c, d) .

Table 1 Optimal dilution/concentration factors (α) for the ELISA with regions $L = [L_-, L_+]$ and $F = [F_-, F_+]$ defined by $(\phi_2, \phi_3) = (-.872, 4.662)$ and various (c, d)

(c, d, λ)	α	$[F_-, F_+]$
(2, 2, 0)	$\{-1.51, 1.51\}$	$[1.64, 7.68]$
(3, 3, 0)	$\{-3.02, 0, 3.02\}$	$[.13, 9.19]$
(6, 6, 0)	$\{-7.55, -4.53, -1.51, 1.51, 4.53\}$	$[-4.40, 13.72]$
(1.67, 2, 1/3)	$\{-1.01, 1.01\}$	$[2.14, 7.18]$
(2.33, 3, 1/3)	$\{-2.01, 0, 2.01\}$	$[1.14, 8.18]$
(4.33, 6, 1/3)	$\{-5.03, -3.02, -1.01, 1.01, 3.02, 5.03\}$	$[-1.88, 11.20]$
(1.33, 2, 2/3)	$\{-.50, .50\}$	$[2.65, 6.67]$
(1.67, 3, 2/3)	$\{-1.01, 0, 1.01\}$	$[2.14, 7.18]$
(2.67, 6, 2/3)	$\{-2.52, -1.51, -.50, .50, 1.51, 2.52\}$	$[.63, 8.69]$

d = number of dilutions; λ = degree of overlap; $L = [3.15, 6.17]$

Recall that at stage 1 of the construction of the design for parallel fitting, the optimal design for classical calibration can be constructed using any desired optimality criterion. We calculated an optimal design for classical calibration using the formulae provided by Rocke and Jones (1997). The input into the formulae were: 4 standards (inclusive of $-\infty$ and ∞ log concentrations), 12 unknowns and $(\phi_1, \phi_2, \phi_3, \phi_4, \sigma) = (.350, -.872, 4.662, .151, 0.045)$. The standards and the unknowns were equally replicated. The calculations obtained the following exact design for classical calibration:

$$\xi_{96} = \left\{ \begin{array}{cccc|cc} -\infty & 3.974 & 6.494 & \infty & S & U \\ 6 & 6 & 6 & 6 & 24 & 72 \end{array} \right\} \quad (23)$$

The design for classical calibration (23) was extended to the designs for parallel fitting (ξ_{96}^p) using the optimal α in Table 1, e.g. to obtain the exact counterpart of design (8).

Table 2 The ranges of the asymptotic variances of the predicted unknown by the parallel fitting method and the classical calibration method (ELISA):

$$(\phi_1, \phi_2, \phi_3, \phi_4, \sigma, K) = (.350, -.872, 4.662, .151, 0.045, 96)$$

(c, d, λ)	$[F_-, F_+]$	Range of the Asymptotic Variance of $\hat{z}^*$	
		Classical Calibration	Parallel Fitting
(2, 2, 0)	[1.64, 7.68]	.010 – .359	.014 – .068
(3, 3, 0)	[.13, 9.19]	.010 – 4.745	.018 – .091
(6, 6, 0)	[–4.40, 13.72]	.010 – 13259.000	.032 – .166
(1.67, 2, 1/3)	[2.14, 7.18]	.010 – .165	.012 – .068
(2.33, 3, 1/3)	[1.14, 8.18]	.010 – .825	.015 – .090
(4.33, 6, 1/3)	[–1.88, 11.20]	.010 – 161.830	.023 – .157
(1.33, 2, 2/3)	[2.65, 6.67]	.010 – .081	.010 – .063
(1.67, 3, 2/3)	[2.14, 7.18]	.010 – .165	.011 – .082
(2.67, 6, 2/3)	[.63, 8.69]	.010 – 1.982	.015 – .136

d = number of dilutions; λ = degree of overlap; $L = [3.15, 6.17]$

Table 2 and Figures 7–12 (in the appendix) compare the robustness and the efficiency of the resultant designs for parallel fitting with that of the corresponding design for classical calibration. For all (c, d) (or for all F), the range of the variance of the predicted unknown by parallel fitting method is smaller than that of the predicted unknown by the classical calibration method. For example (see Table 2), for $(c, d) = (6, 6)$ (or $F = [-4.40, 13.72]$) the variance of the predicted unknown by the classical calibration method varies from 0.010 to 13251 whereas that by the parallel fitting method varies from 0.032 to 0.166. This observation means the parallel fitting method is more robust for predicting a wide range of the unknowns than the classical

calibration method. However, for all (c, d) (or for all F) the classical calibration method is more efficient for predicting moderate unknowns than the parallel fitting method. For example, see Figures 7–12 in the appendix. The inefficiency of the parallel fitting method in predicting moderate unknowns is most pronounced when F is too wide relative to L , or equivalently, when c and d are large. Hence, in practice (c, d) should be chosen to compromise the robustness and the efficiency of predicting both moderate and extreme unknowns.

4. Conclusion

This paper has shown that the parallel fitting method is a formidable competitor of the classical calibration method in that the method is more robust and reasonably efficient for predicting both moderate and extreme unknowns. As is usual with robust statistical methods, the parallel fitting method compromises the efficiency of predicting moderate unknowns with the efficiency and robustness of predicting extreme unknowns. It is shown how to compromise the robustness and the efficiency of the parallel fitting method, relative to the classical calibration method, by design. The compromise is possible due to the fact that the designs for the parallel fitting method are easily constructed by combining designs for classical calibration with hand calculated optimal dilution/concentration factors. Unfortunately it is not easy to analytically study the behaviour of bias and the variance of the predicted unknown (by the parallel fitting method) as a function of all the design variables, hence this paper has resorted to numerical methods.

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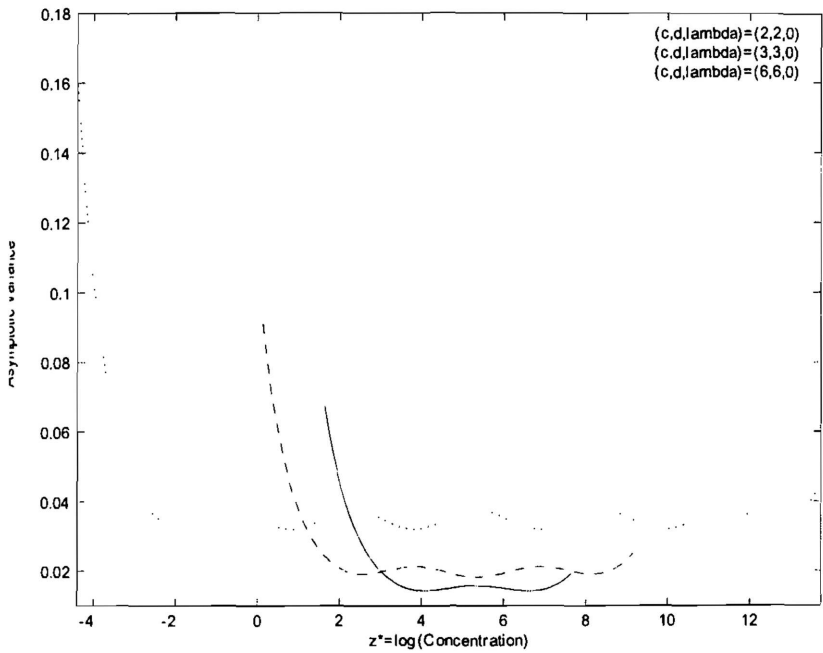


Figure 7 The asymptotic variances of the predicted unknown by the parallel fitting method (ELISA): $(c, d, \lambda) = (c, d, \lambda)$

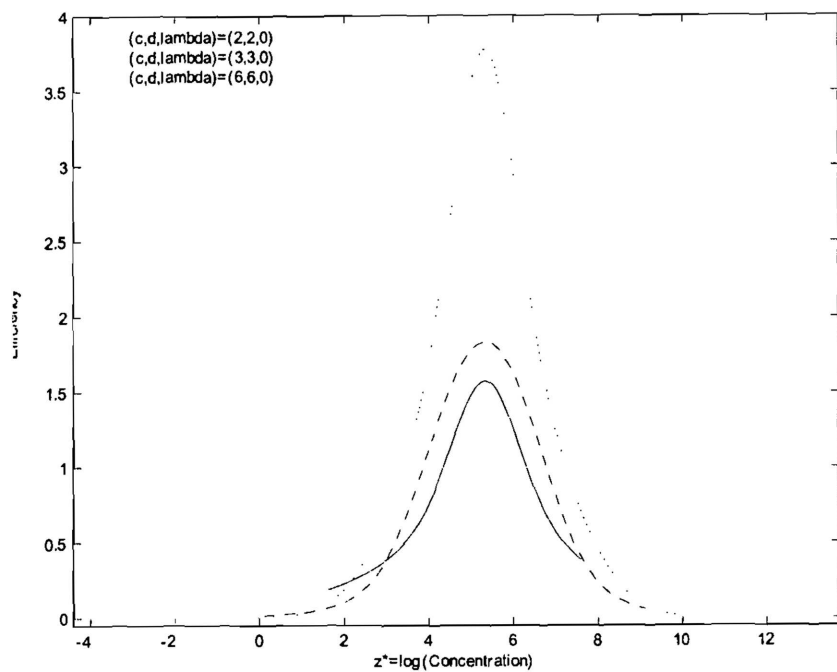


Figure 8 The efficiency of the parallel fitting method relative to the classical calibration method $(= V(\phi, z^*, \xi_{96}^p) / V(\phi, z^*, \xi_{96}))$ (ELISA): (c, d, λ)
 $= (c, d, \lambda)$

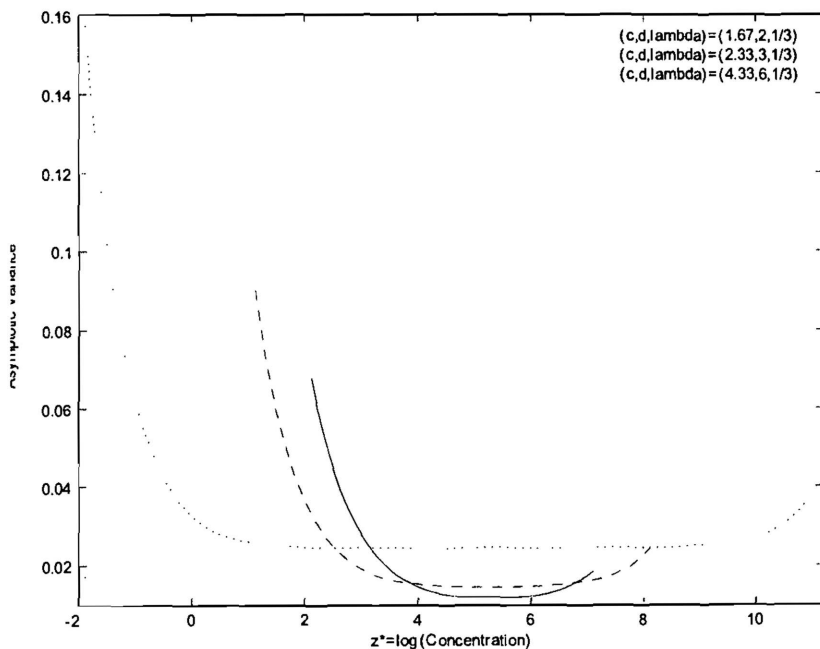


Figure 9 The asymptotic variances of the predicted unknown by the parallel fitting method and the classical calibration method (ELISA): $(c, d, \lambda) = (c, d, \lambda)$

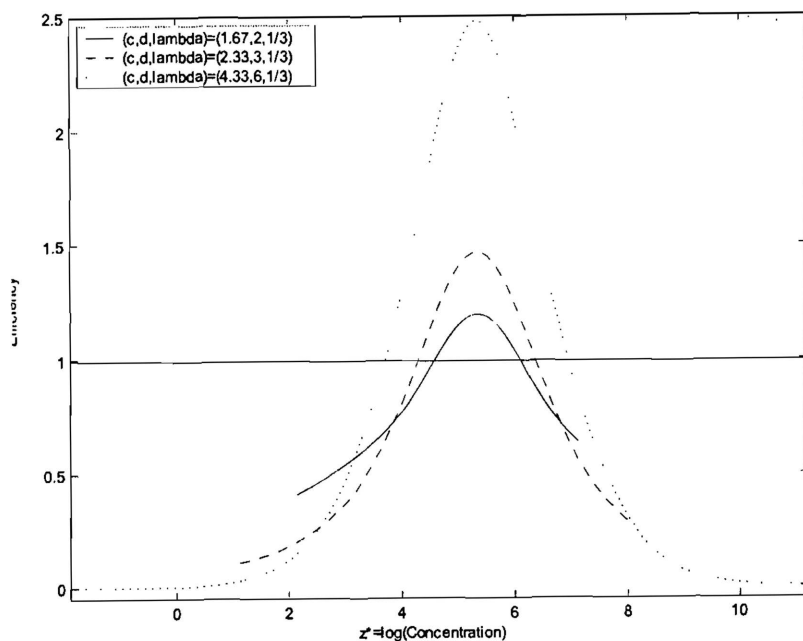


Figure 10 The efficiency of the parallel fitting method relative to the classical calibration method $= V(\phi, z^*, \xi_{96}^p) / V(\phi, z^*, \xi_{96})$ (ELISA):
 $(c, d, \lambda) = (c, d, \lambda)$

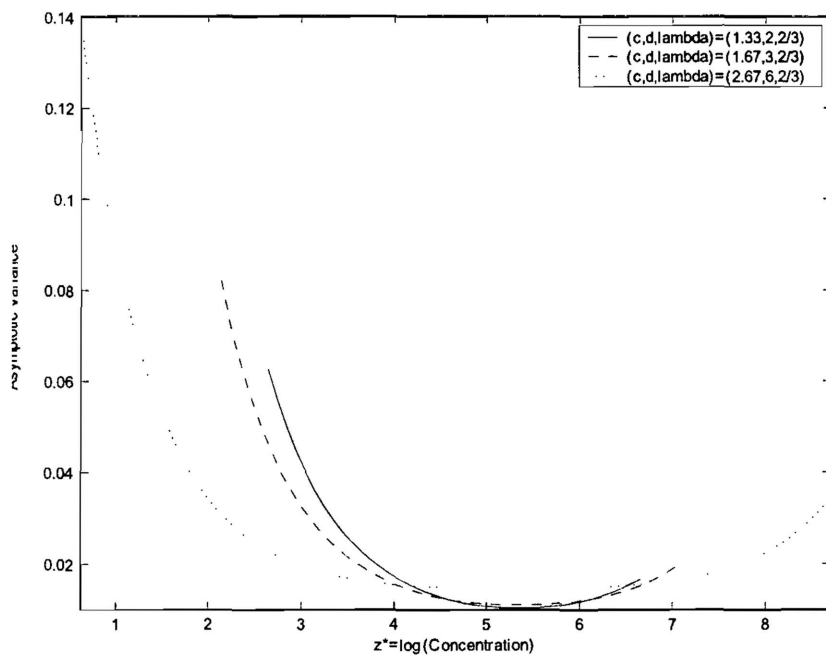


Figure 11 The asymptotic variances of the predicted unknown by the parallel fitting method and the classical calibration method (ELISA):
 $(c, d, \lambda) = (c, d, \lambda)$

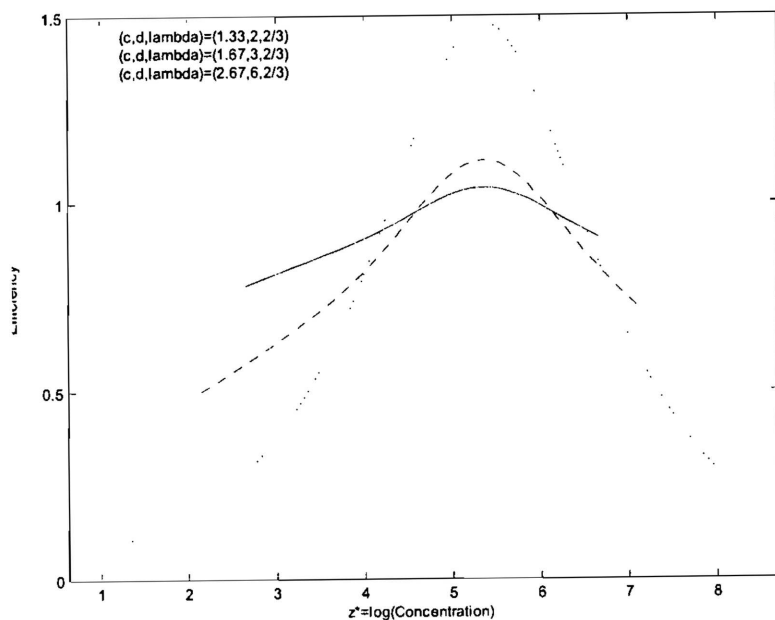


Figure 12 The efficiency of the parallel fitting method relative to the classical calibration method $(= V(\phi, z^*, \xi_{96}^p) / V(\phi, z^*, \xi_{96}))$ (ELISA):
 $(c, d, \lambda) = (c, d, \lambda)$